

A1 Receive-Coil Amplifier

Hearing-Loop Receiver — Structured Design Review

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Outline

1. Specification & design environment
2. Transfer function & feedback configuration
3. First stage: noise design (g_m , C_{iss} , EKV)
4. Single stage & loop-gain check
5. Dual stage (PMOS pair + NMOS CS)
6. EKV model & pole-zero analysis
7. Frequency compensation (MFM)
8. Conclusions

What the review wants

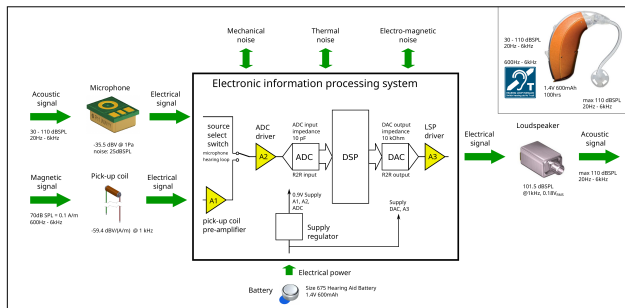
A structured, *top-down* story: set the budget first, generate options, then justify every component before placing it — “*replace the how with the why.*”

Part 1

Specification & design environment

The hearing loop — and where A1 lives

Application diagram



Application diagram — EE4109 HearingLoopIntro

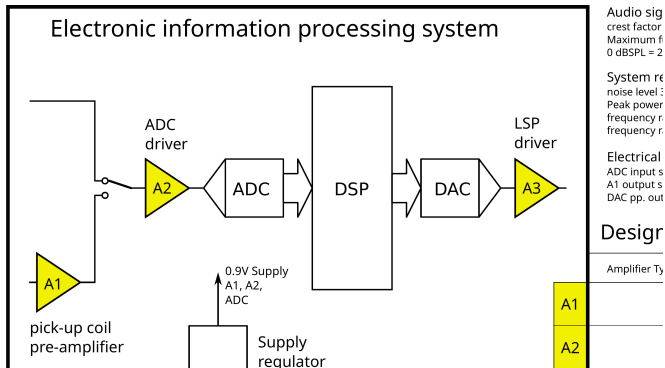
A magnetic *induction loop* broadcasts the audio; a hearing aid picks it up with a coil.

- **A1** = receive-coil pre-amplifier.
- Feeds A2 (ADC driver) → ADC → DSP.

Why

Before any transistor: understand the *environment* and what A1 connects to. That sets the whole specification.

A1 in the signal chain

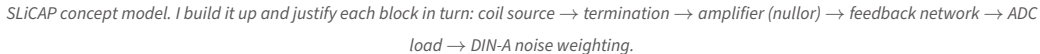


Electronic information processing system — A1 is the pick-up-coil pre-amplifier

- **Input:** pick-up coil.
- **Output:** source-select switch (mic / loop), then the ADC driver.
- Single 0.9 V supply; 1 mW total power budget.

Three questions first

1. What environment?
2. How does it work?
3. What is the budget?



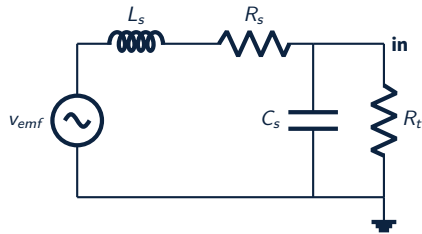
Representation 1 — the pick-up coil (a *differentiator*)

A current in the loop makes a magnetic field; the coil sees a voltage proportional to the *rate of change* of flux:

$$v_{coil}(t) = -N \frac{d\Phi}{dt} \propto \frac{dB}{dt}$$

Small-signal model of the coil + cable:

- series $L_s = 120 \text{ mH}$, $R_s = 875 \Omega$
- parasitic $C_s \approx 9.4 \text{ pF}$
- termination $R_t \approx 80 \text{ k}\Omega$

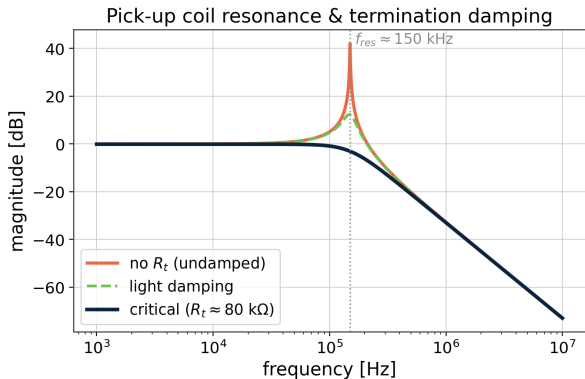


Pick-up coil two-port

Why

An **integrator** A1 undoes this differentiation
 $\Rightarrow$ flat audio response.

Representation 2 — the termination resistor



L_s and C_s resonate at

$$f_{res} = \frac{1}{2\pi\sqrt{L_s C_s}} \approx 150 \text{ kHz}.$$

Undamped, this peak rings and overloads the input.

- Add R_t across the input.
- Pick R_t for *critical* damping, $Q = \frac{1}{\sqrt{2}}$.

What the review wants

“If you add a component, have a *reason*.” R_t exists only to damp the LC resonance — and it is then a noise source we must budget.

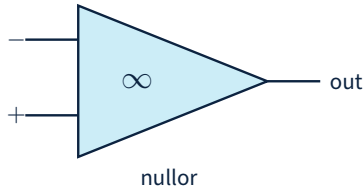
Representation 3 – the ideal amplifier (nullor)

I model the controller as a **nullor**: the ideal element with infinite gain.

- Input (nullator): $v = 0, i = 0$.
- Output (norator): v, i free.

With a nullor the transfer is set *only* by the passive feedback:

$$A_{\infty} = \frac{1}{s R_f C_i} = \frac{62.4 \times 10^3}{s}$$



Why

The nullor separates the *ideal target* from the implementation. Real transistors come later.

Representation 4 — the output (switch, ADC, C_L)

- **Source-select switch** (microphone / hearing loop).
 - ▶ requires $V_{mic} = V_{loop}$ at the reference frequency $f_{ref} = 1$ kHz.
- **ADC input** behaves as a voltage amplifier:
 $Z_{in} \rightarrow \infty$, draws no current.
- Only loading left = parasitic $C_L = 10$ pF .

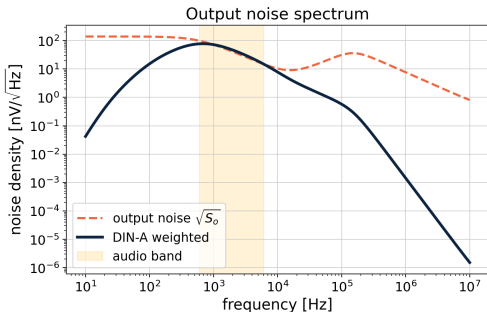
Consequence

A1 is a **voltage-voltage** amplifier driving a capacitive load — not a power stage.

Why

Matching $V_{mic} = V_{loop}$ at f_{ref} is what fixes the required gain magnitude.

Representation 5 — the noise sources



Each resistor adds thermal noise $\overline{v^2} = 4kTR$; the input device adds v_n, i_n (the *noisy nullor*).

- Budget $B_{n1} = 0.4$ to R_s, R_t, R_f .
- The rest goes to the input transistor $\Rightarrow$ this sets g_m .
- Output-noise RMS $\approx 7.5 \mu\text{V}$ (1 Hz–100 kHz), within the $10.6 \mu\text{V}$ DIN-A budget. ✓

What the review wants

Every noise source is traced to the budget — nothing is added without accounting for it.

How should A1 work?

1. A1 is a **voltage–voltage amplifier**.
2. Its transfer is an **integrator** (cancels the coil's differentiation):

$$H(s) = \frac{V_o}{V_i} = \frac{1/\tau_i}{s} = \frac{62.4 \times 10^3}{s} \text{ V/V}, \quad \tau_i = 16.06 \mu\text{s}$$

3. Gain magnitude set so $V_{mic} = V_{coil}$ at f_{ref} .
4. **Noise:** weighted output noise must not exceed the microphone's floor (≤ 30 dB SPL).

What the review wants

Note the *order*: transfer first, *then* noise. Keep it a V–V amplifier throughout — do not silently switch to a transimpedance amplifier.

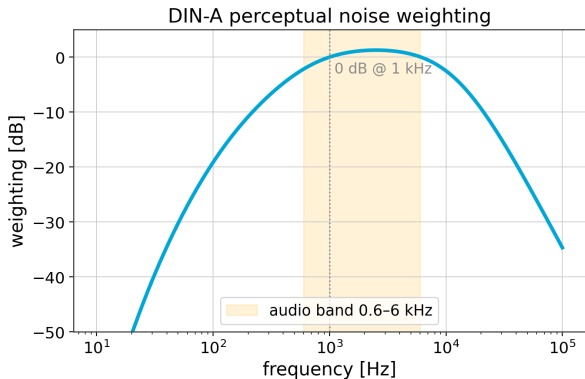
Specification & budget

Quantity	Value	Sets...
Transfer (integrator)	$62.4 \times 10^3 / s$ ($\tau_i = 16.06 \mu s$)	gain / $R_f C_i$
Audio band	$f_{min} = 600 \text{ Hz} \dots f_{max} = 6 \text{ kHz}$	bypass & bandwidth
Output noise (DIN-A)	$\leq 10.6 \mu V$ RMS ($\approx 30 \text{ dB SPL}$)	input stage
Load	$C_L = 10 \text{ pF}$	stability / drive
Supply / power	$1.1 \text{ V to } 1.3 \text{ V} / \leq 1 \text{ mW}$	currents
Coil	$L_s = 120 \text{ mH}, R_s = 875 \Omega, C_s = 9.4 \text{ pF}$	source noise / R_t

Why

Every later decision is traced back to a line in this table. The amplifier is graded on *this reasoning*, not on whether the final numbers are perfect.

Noise is judged by ear: DIN-A weighting



The ear is not flat. The output noise is weighted by the standard **DIN-A** curve before integrating to an RMS value:

$$\sigma_o^2 = \int_0^\infty |A(f)|^2 S_o(f) df.$$

- 0 dB at 1 kHz.
- Strong roll-off below 600 Hz / above a few kHz.

Why

Weighting tightens the budget exactly where hearing is sensitive — so the budget is perceptually meaningful.

Part 2

Transfer function & feedback configuration

First generate the options — *then* choose

Option A — voltage-voltage

(integrator in feedback)

- $Z_1 = \text{capacitor}$, $Z_2 = \text{resistor}$.
- Coil seen as a voltage source.
- Coil R_s only enters *noise*.

Option B — current-voltage

(transimpedance, inductor integrates)

- $Z_2 = \text{inductor}$, $Z_1 = \text{resistor}$.
- Coil R_s now enters the *transfer*.
- Large inductor $\Rightarrow$ area / noise.

What the review wants

The prereview lesson: **pick one and stay consistent**. Mixing a V-V amplifier on one slide with a transimpedance amplifier on the next makes the plots meaningless.

Decision

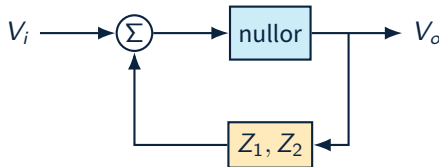
Choose **Option A: voltage-voltage integrator**. Practical components, and R_s stays out of the transfer.

Why feedback? The asymptotic-gain model

Feedback fixes the transfer with a passive network, so accuracy follows passive components — not the transistor.

$$A_f = A_\infty \underbrace{\frac{-L}{1-L}}_{\text{servo}} + \underbrace{\frac{1}{1-L}}_{\text{direct}}$$

- A_∞ : ideal (nullor) transfer = $1/(sR_f C_i)$.
- L : loop gain (return ratio).
- servo $\rightarrow 1$ when $|L| \gg 1 \Rightarrow A_f \rightarrow A_\infty$.



What the review wants

In the review you must *explain every curve* of this model — gain, asymptotic, loop gain, servo, direct.

Sizing the feedback network

Bound R_f from two budgets:

- **Noise** budget ($B_{n1} = 0.4$) $\Rightarrow$ *upper* bound
 $R_{f,max} = 7.5 \text{ k}\Omega$.
- **Power** budget ($B_{i1} = 0.1$) $\Rightarrow$ *lower* bound
 $R_{f,min} = 1.4 \text{ k}\Omega$.

Then set C_i from the integrator constant and add a bypass R_b for the DC path:

$$C_i = \frac{\tau_i}{R_f}, \quad R_b = \frac{1}{2\pi C_i f_{min}}.$$

$$R_f = 3.24 \text{ k}\Omega$$

$$C_i = 4.96 \text{ nF} \quad R_b = 53.5 \text{ k}\Omega$$

Why

$R_b \parallel C_i$ sets the lower corner at $f_{min} = 600 \text{ Hz}$ and gives the DC bias a path — one resistor, two justified jobs.

Part 3

First stage: noise design

The first stage sets the input-referred noise

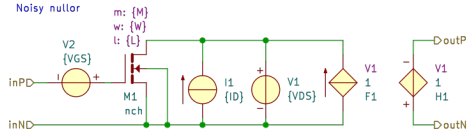
NMOS noisy nullor circuit

Models a nullor with NMOS input-referred noise sources
 NMOS parameters: W, L, M, ID, VGS, VDS (ID>0, VGS>0, VDS>0)
 Put the library directive in the top-level circuit

A1

```
.subckt NoisyNullorN outP outN inP inN W=220n L=180n M=1 ID=1u VGS=0.6 VDS=0.9
```

Noisy nullor



Noisy nullor: ideal nullor + the input device's v_n , i_n

Replace the controller by a **noisy nullor**: an ideal nullor plus the input transistor's noise sources.

- Later stages add almost nothing (divided by the first-stage gain).
- So the whole noise budget is spent on *stage 1*.

Why

This is why we design the first stage *from the noise spec* — before sizing anything else.

The knobs: g_m , c_{iss} , inversion

Input-referred thermal noise of a CS stage:

$$\overline{v_n^2} = \frac{4kT\gamma}{g_m} [\text{V}^2/\text{Hz}]$$

- More $g_m \Rightarrow$ less noise (more current for less voltage).
- But wider device $\Rightarrow$ more $c_{iss} \Rightarrow$ more input loading.
- **Weak inversion** gives the largest g_m/I_D .

The trade-off

Minimise the product $g_m \cdot c_{iss}$: enough g_m for the noise budget, lowest c_{iss} for the source.

What the review wants

“Replace the how with the why”: we choose weak inversion *because* it maximises transconductance efficiency for the noise budget.

Which stage? CS vs CG vs CD

	Gain	Chain (ABCD)	Role
CS	high	smallest entries	most <i>nullor-like</i>
CG	≈ 1 (current)	—	current follower
CD	≈ 1 (voltage)	—	voltage follower

- The CS chain matrix has the *smallest* elements $\Rightarrow$ it behaves most like a nullor, so following stages barely add noise.
- CG and CD are *followers* — no voltage gain to suppress later noise.

First stage = **common-source**, in saturation.

From budget to device parameters

1. Find the output noise spectrum $S_o(f)$.
2. Integrate (DIN-A weighted) $\rightarrow$ variance $\sigma_o^2 = (B_{n2} V_{o,n})^2$.
3. Express g_m as a function of c_{iss} .
4. Lowest feasible $c_{iss} \rightarrow g_m, I_D$, then W, L via the **EKV** model.

PMOS input ($g_m \cdot c_{iss}$ optimum):

$$g_m = 73 \mu\text{S}, W = 19.1 \mu\text{m}$$

$$L = 0.72 \mu\text{m}, I_D = 8.1 \mu\text{A}$$

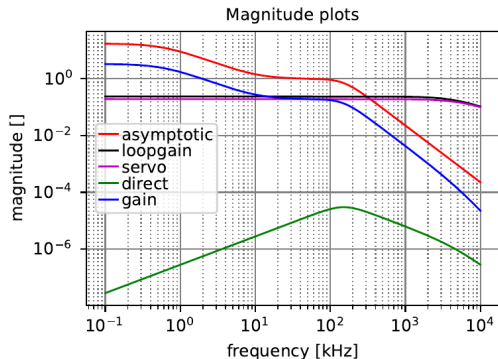
Why

SLiCAP gives the symbolic spectrum; the EKV model turns the budget directly into W, L, I_D . ngspice then verifies the numbers.

Part 4

Single stage & loop-gain check

Is a single stage enough?



Single-stage feedback model — the loop gain L stays below 0 dB.

A single CS has only 3 usable terminals $\Rightarrow$ a **differential pair** ($\approx 4\times$ current and area).

- Current-drive into C_L : ✓
- Voltage headroom on 0.9 V: ✓
- Loop gain $|L_{DC}| < 1$ across the band: ✗

Verdict

$|L| < 1 \Rightarrow$ the servo cannot enforce the transfer
 $\Rightarrow$ **poor accuracy**. Go to a **dual stage**.

Part 5

Dual stage (PMOS pair + NMOS CS)

Why a dual stage?

The five costs trade against each other:
noise, power, accuracy, stability, area.

- The **first stage dominates noise**.
- The **second stage dominates loop gain** (accuracy).

Split the job: optimise each stage for one thing.

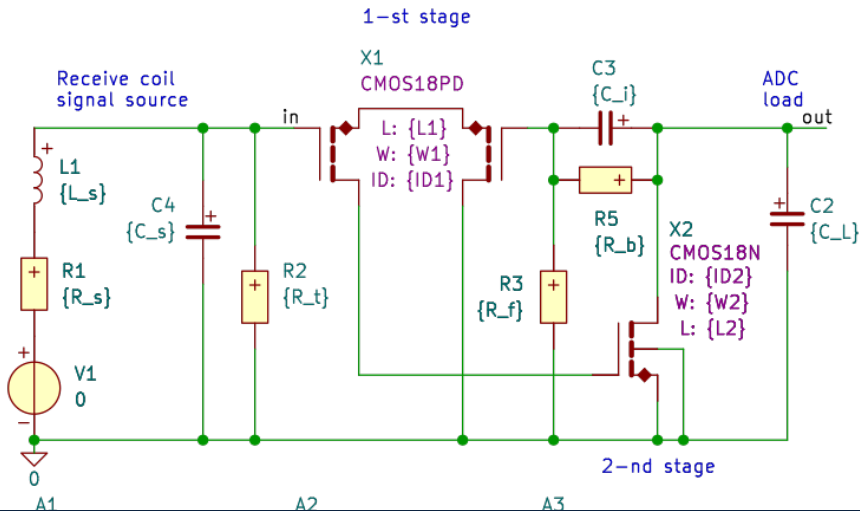
Stage 1: **PMOS differential pair** — low noise.

Stage 2: **NMOS common-source** — high loop gain.

Why

Two stages give enough loop gain for accuracy while keeping the input noise low — the single stage could not do both.

The dual-stage topology



First stage: differential pair vs single CS

Differential pair

- + lower noise
- + high CMRR
- + matches the coil's 2-terminal input
- more area
- higher complexity

Single-ended CS

- + less area
- + lower power
- + simple
- allows CM interference
- higher noise

The coil is a floating two-terminal source $\Rightarrow$ the **differential pair** wins on noise and common-mode rejection.

Second stage: NMOS common-source

Chosen for what the first stage is *not* optimised for:

- high loop gain $\Rightarrow$ accuracy;
- high current-drive into C_L ;
- low power, small area, simple.

Minimum length $L = 0.18 \mu\text{m}$ for speed; width traded for g_m/I_D .

NMOS CS output stage:

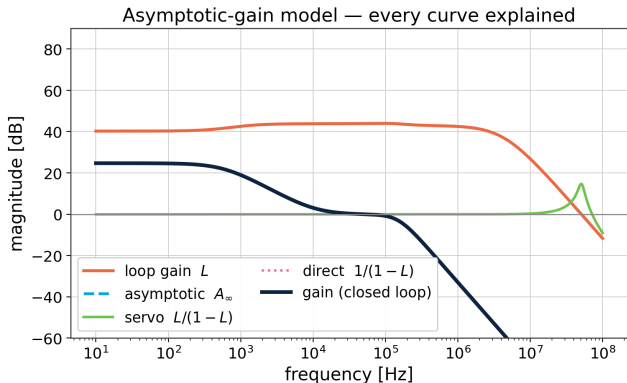
$$W = 10 \mu\text{m}, L = 0.18 \mu\text{m}$$

$$I_D = 16.5 \mu\text{A}$$

Why

Drive current $I_D \approx V_{i,pp}/2R_f$ must charge C_L at the full-power frequency.

Asymptotic-gain model — every curve explained



- **gain**: the closed-loop transfer we deliver.
- **asymptotic** A_∞ : ideal nullor target ($1/sR_f C_i$).
- **loop gain** L : ≈ 45 dB at DC — the “budget” for accuracy.
- **servo** $L/(1-L)$: $\rightarrow 1$ while $|L| \gg 1$, so gain $\rightarrow A_\infty$.
- **direct** $1/(1-L)$: feed-through; negligible here.

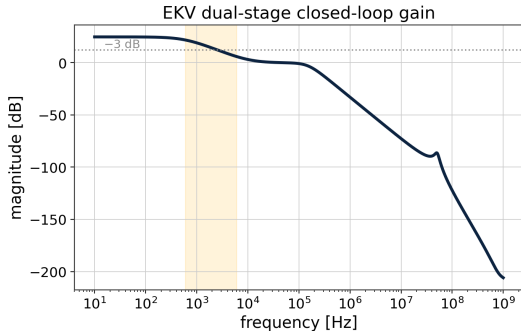
What the review wants

Gain follows A_∞ exactly where the servo ≈ 1 ; it departs once $|L|$ falls.

Part 6

EKV model & pole-zero analysis

From behavioural to the EKV transistor model

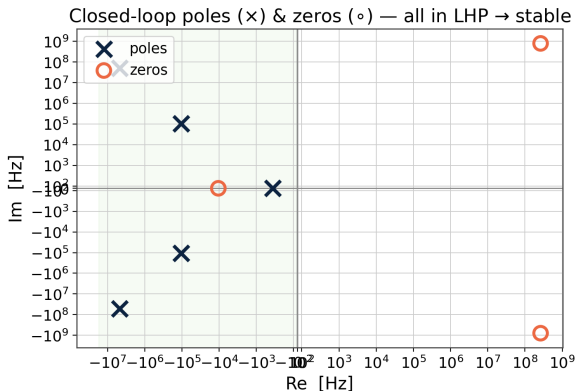


Swap the ideal devices for the **EKV** small-signal model:

- realistic g_m , g_o and *all* parasitic capacitances;
- far more accurate gain and phase;
- reveals the real poles & zeros.

The passband matches A_∞ ; the high-frequency behaviour is now trustworthy.

Pole-zero analysis



5 poles, all with $\text{Re} < 0 \Rightarrow$ stable.

- ≈ 600 Hz: dominant (integrator + bypass).
- ≈ 150 kHz pair: $Q \approx 0.71$ (well damped).
- ≈ 51 MHz pair: $Q \approx 5.4$ — severely **underdamped** $\Rightarrow$ peaking.

What the review wants

The gain plot follows the poles and zeros: the high- Q pair is exactly the bump we must explain — and then remove.

Part 7

Frequency compensation (MFM)

Target: maximally-flat magnitude (MFM)

A pole pair with $Q > \frac{1}{\sqrt{2}}$ peaks and rings. **MFM** (Butterworth) sets the dominant closed-loop pair to

$$Q = \frac{1}{\sqrt{2}} \approx 0.707.$$

- No peaking, fastest flat response.
- Comfortable phase margin.

How to move a pole pair?

Add a zero to the loop gain near the offending poles — it injects phase lead and lowers the effective Q .

Why

We shape *stability*, not just gain — a Bode plot is the 0-axis slice of the Nyquist plot.

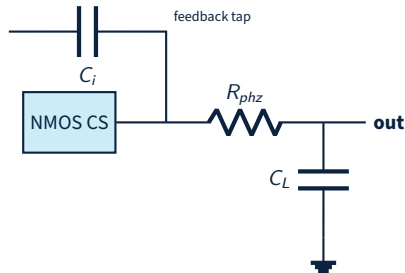
The phantom-zero method

Put a resistor R_{phz} in series with the load capacitor. It creates a zero in the *loop gain* without a real pole in the signal path:

$$z = -\frac{1}{R_{phz} C_L}$$

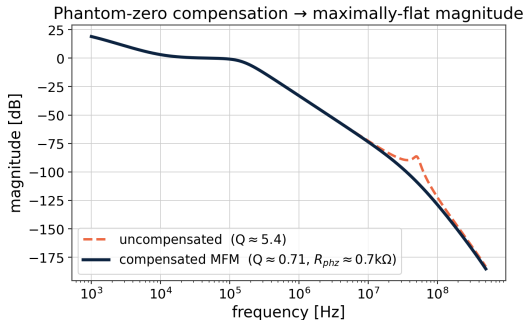
Why this one?

- no extra noise source in the signal path;
- negligible power and area;
- no added energy-storage (no compensation cap).



R_{phz} in series with the output; feedback is taken before it, so the zero appears only in the loop gain.

Compensation result



$$R_{phz} \approx 0.70\text{ k}\Omega$$

$$Q : 5.4 \rightarrow 0.71 \text{ (MFM)}$$

- Peak removed; passband untouched.
- All poles remain in the left half-plane $\Rightarrow$ stable.
- Cost: one resistor.

Why

The phantom zero buys maximally-flat magnitude for essentially zero noise/power penalty.

Part 8

Conclusions

Conclusions

The structured path, end to end:

1. Specification & budget from the environment.
2. V-V integrator chosen *after* listing options.
3. Noise budget $\rightarrow$ first-stage $g_m, c_{iss} \rightarrow W, L, I_D$.
4. Single stage fails accuracy $\rightarrow$ dual stage.
5. PMOS pair (noise) + NMOS CS (loop gain).
6. EKV pole-zero: stable, one high- Q pair.
7. Phantom zero $\rightarrow$ maximally-flat magnitude.

What the review wants

The grade is in the *reasoning*: right order, options before choices, every curve and component justified — the how replaced by the why.

$$H(s) = \frac{62.4 \times 10^3}{s}, \text{ noise} \\ \leq 10.6 \mu\text{V, stable, MFM.}$$

Thank you

Questions & discussion